

Jaxon Doolittle

Software and data infrastructure for scientific and engineering projects

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What I do

I do research and experimental work for NASA, but **I am an engineer at the core**. What I am hired for is the software and data infrastructure underneath the science: processing pipelines, monitoring and alerting, build and test automation, version control, and the reproducibility and provenance layer beneath it. The recurring problem is a system that fails quietly, produces numbers nobody can reconstruct, or only runs because one person remembers how it works.

Selected results

- **NASA-funded space weather forecasting pipeline, NASA Pleiades supercomputer** (CLEAR Center, University of Michigan). Led the compute and reliability overhaul in support of Artemis.
- Closed monitoring gaps that let production failures run undetected for up to five days. Added health checks and same-hour alerting, roughly a **120:1 cut in worst-case detection time**.
- Resolved more than ten reliability and maintainability defects across a six-phase remediation: an unmonitored single point of failure, a job-dispatch race condition, production scripts outside version control.
- Built a real-time forecast correction layer on the site physics model, **cutting RMSE 53 percent** (134 to 63 km/s) and turning disturbed-period tracking from net-negative correlation into a predictive signal.
- **Enterprise security awareness platform**, roughly two and a half years. Full-stack Python platform running AI voice phishing simulations for multiple client companies, with an ML pipeline scoring employee vulnerability and clustering behavioral response archetypes.

Services and typical scope

Service	Typical scope	Turnaround	Fee
Reliability and monitoring build-out	Find where the system fails silently, then fix it. Health checks, alerting, stall detection, auto-restart, single points of failure.	3 to 6 weeks	\$6,000 to \$20,000
Reproducibility and data provenance	Provenance capture, versioning, environment pinning, documented formats, and a working path for an outside party to rerun the results.	3 to 5 weeks	\$4,000 to \$12,000
Software sustainability and test infrastructure	Build and test automation, regression coverage on numerical output, dependency hygiene, getting code out of one person's head and into a repository.	4 to 8 weeks	\$8,000 to \$25,000
Security awareness and social engineering resilience	Phishing and voice phishing simulation, employee risk scoring, awareness program design. See the scope note below for what this does and does not cover.	2 to 6 weeks	\$3,000 to \$15,000
Proposal section support	Data management, software sustainability and computational approach sections written to the agency solicitation, plus one revision round.	1 to 2 weeks	\$1,500 to \$4,000

Fixed fee against a written scope, never hourly. Ranges only; every engagement is quoted specifically.

Scope note on cybersecurity

What I do: the human attack surface. Phishing and voice phishing simulation, behavioral risk scoring, security awareness program design, and the engineering side of technical controls such as audit logging, configuration management and dependency hygiene. **What I do not do:** CMMC or NIST SP 800-171 compliance work. I am not a registered practitioner or assessor, I do not write System Security Plans or POA&Ms, and I would not take on assessment readiness. A company that needs certification should hire someone who does that specifically.

Background and how engagements start

MS in Computer Science in progress, Georgia Southwestern State University. BAS Information Technology. Active NASA Advanced Supercomputing credentials, with current work running on Pleiades. Engagements begin with a short call, then a written scope with a fixed fee and a delivery date before any work starts. For a first engagement I am glad to begin with one narrow deliverable at the low end of the range, so the work product speaks before anything larger is committed.

The University of Michigan and NASA are named to describe where my work has been performed. Neither is affiliated with, nor endorses, this independent practice, and no work product belonging to either is offered or shared through it.